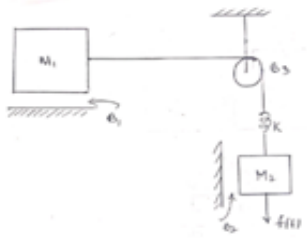


CONTROL SYSTEMS – UE24EE241B

Sample Questions

Analogous Systems	
1.	Analogous systems are those which have: A. Same physical components B. Same mathematical model C. Same energy source D. Same output only
2.	Which of the following is an example of analogous systems? A. Electrical and thermal systems B. Mechanical and hydraulic systems C. Electrical and mechanical systems D. All of the above
3.	In force–voltage analogy, the mechanical impedance $Z_m(s) = Ms + B + \frac{K}{s}$ is electrically analogous to:
4.	The electrical analog of a mechanical system has an inductor of value 2 H, resistor of 4 Ω , and capacitor of 0.5 F in series. Using force–voltage analogy, the mechanical parameters are:
5.	A mechanical impedance is given by: $Z_m(s) = 5s + 10 + \frac{20}{s}$ The equivalent electrical impedance (force–voltage analogy) is:
6.	A mechanical system is governed by: $F(s) = (6s^2 + 3s + 12)X(s)$ Using force–voltage analogy, find the electrical impedance $Z(s)$.
7.	A rotational mechanical system has: • Moment of inertia $J = 5$ • Friction coefficient $B = 10$ • Torsional spring constant $K = 20$ Using torque–voltage analogy, find the electrical circuit parameters.

8.	The electrical analog of a mechanical system has: • $L = 3 \text{ H}$ • $R = 6 \Omega$ • $C = 0.25 \text{ F}$ Using force–voltage analogy, find the natural frequency of the mechanical system.
9.	A mechanical system has transfer function: $\frac{X(s)}{F(s)} = \frac{1}{s(2s + 4)}$ What are the analogous electrical elements in f-v analogy?
10.	Identify the number of displacements for the given mechanical system  ☐ 4 ☐ 2 ☐ 3 ☐ 1

Date: 23-01-26

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(Anchor)